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OPM401: IT Project Management

Machine Learning Project

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Project Management Concepts: Machine Learning Project

1. Definition
Time-bound, unique endeavour

- Predict student dropout.
- Using Python & ML algorithms
- Has a sponsor or customer
- Degree of uncertainty

2. Attributes
7 key project characteristics

- Clear objective
- Interdependent activities
- Various resources (people, data, tools)
- Specific time frame.

3. Objective (SMART)
What the project aims to achieve

- Specific: Predict student dropout
- Measurable: 85% model accuracy
- Achievable: Python / Scikit-learn
- Time Bound: 6 months

4. Deliverables
Tangible outputs of the project

- Train ML model
- Data preparation report
- Accuracy and evaluation report
- Deploy prediction system
- Final user guide and documentation.

5. Constraints
Scope, Time, Cost, Quantity

- Scope: ML model only (not full system)
- Time: Deadline - 8 May 2026
- Cost: Limited budget for cloud tools
- Quantity: Min 85% accuracy requirement
- Resources: Team of 5 members

6. Project Life Cycle
5 Phases Of the ML project

- Initiating: Define needs and charter
- Planning: Schedule, budget, risk
- Executing: Build and train ML mode
- Monitoring: Track and manage changes
- Closing: Deliver model and lessons learned

7. Benefits of PM
Why use project management?

- Keeps project on time and budget
- Clear communication and accountability
- Early identification of risk
- Structured, organised approach
- Higher quality of ML deliverables

8. PMBOK Knowledge Areas
Areas applied to ML project

- Integration: Coordinate all ML tasks
- Scope: Define what model will/won't do
- Time: Schedule data collection and training
- Cost: Budget for tools and cloud
- Risk: Manage data quantity and model bias

Project Manager:

A project manager is the person who plans, organizes, and controls the project to meet its goal within the specified scope, time, and budget. To successfully perform these roles, a project manager needs a mix of technical and soft skills, including leadership, communication, and change management.

Project Manager Profile:

Name: Rohit Aroonslam

Project: Machine Learning Project Student Dropout System

Organisation: School of IT - Nelson Mandela University

Roles: Team organizer. Project planner, Project supervisor

Responsibilities: *Planning*, *Organising*, and *Controlling*. A good project manager needs to be able to plan accordingly and strategically, they also need to be able to organize the project team (through selection, and assigning responsibility) by tasks and organize task due dates in order of importance/priority, for a smoother transition into each phase of the project, and finally, a good project manager needs to be able to control their surroundings, control the team members and take action if the project tasks start to deviate from the critical path towards success and the projects overall goal.

Skill	Application to ML Project
Leadership	Guiding the team through complex ML model development
Communication	Reporting progress to the supervisor and keeping the team aligned
Delegation	Assigning data prep, model training, and testing to the right members
Change management	Handling scope changes if model requirements shift mid-project
Problem-solving	Resolving issues like poor data quality or model underperformance
Technical awareness	Understanding ML concepts enough to make informed decisions



The Project Team:

A project team is a group of individuals working interdependently to achieve the project objective. Teams are formed based on the expertise, experience, and skills required, and should be kept as small as feasible throughout the project. Effective project teams are characterised by a clear understanding of the project objective, clear role expectations, a results orientation, a high degree of collaboration, and a high level of trust. Teams evolve through Tuckman's four stages of development - Forming, Storming, Norming, and Performing - and the project manager must actively support the team through each stage. Barriers to team effectiveness include unclear objectives, poor communication, lack of commitment, and dysfunctional behaviour. Diversity within the team, including differences in age, gender, ethnicity, and background, is a strength that enhances creativity and problem-solving when managed respectfully and inclusively.

Role	Team Member	Responsibilities
Project Manager	Member 1	Overall planning, controlling, and communication with the sponsor
Data Engineer	Member 2	Data collection, cleaning, and preparation for the ML model
ML Developer	Member 3	Building, training, and tuning the ML model
Software Developer	Member 4	Deploying the model into a usable application/system
Quality Assurance Tester	Member 5	Testing model accuracy, writing evaluation report

Stage	What Happens	ML Project Example
Forming	Team gets acquainted, low productivity, dependent on direction	Team meets for kick-off, roles assigned, project explained
Storming	Conflict arises, personalities clash, and frustration builds	Disagreements over which ML algorithm to use or data sources
Norming	Conflicts resolved, trust builds, team-led decisions	Team agrees on approach, and collaboration improves
Performing	High performance, open communication, strong output	Model training runs smoothly; team self-manages effectively

Diversity Considerations: Age/Generation - Team members from different year groups bring different perspectives on technology use as well as problem-solving techniques/approaches. Gender - A gender diverse team brings complementary leadership styles and unique points of view on the same topic. Ethnicity/Ancestry - Different cultural backgrounds enrich discussion and interest in one another. Academic background - Members with backgrounds in IT, statistics, and business bring different analytical approaches to the ML problem. Skill Level - Mix of experienced and junior members encourages mentoring, knowledge sharing, and teamwork.

Team Development Stages:

Initial Stage: Team members get acquainted with each other. Members have positive expectations of the group. There is low productivity as this is only the “forming” stage.

Second Stage: Challenging tasks are assigned and begun. The team’s limits are tested. Conflict arises amongst group members. Individuality is expressed through work and during periods of low morale.

Third Stage: Stable relationships are formed after resolving conflict. Open and welcome feedback. Team-led decision making.

Fourth Stage: Performance amongst members is at its peak. Communication is open. The team feels empowered through a collaboration and mentorship style of leadership.

Project Communication and Documentation:

Communication is the key to project success, as it facilitates coordination and engagement of stakeholders. A communication plan establishes what information is exchanged with whom, when, and through which mediums, as well as specifies meetings, reports, and tools. Project managers should have good communication skills since they spend approximately 90% of their time communicating. (PMI, 2025).

1. Stakeholder Communication Table:

Stakeholder	Information Required	Frequency	Method	Responsible
Supervisor	Progress, risks, and budget status	Bi-weekly	Report and Email	Project Manager
Team	Task updates, decisions, issues	Weekly	Team Meeting	Project Manager
Data Engineer	Data requirements, deadlines	As needed	Messages	ML Developer
QA Tester	Testing criteria and outputs	Testing phase	Email and Docs	ML Developer

2. Meeting Plan:

- **Kick–off Meeting (Once)**: Align team with goals, roles, and plan.
- **Weekly Status**: Monitors progress, delegates tasks.
- **Problem – Solving**: Address technical issues.
- **Final Review (Once)**: Confirm deliverables with supervisor.

3. Reporting:

- **Progress Report (Bi-weekly)**: Completed tasks, issues, and next steps.
- **Risk Report (Monthly)**: New/escalated risks.
- **Final Report (End)**: Outcomes and lessons learned.

4. Tools:

- **Google Drive**: Documents
- **MS Teams / Zoom**: Meetings
- **WhatsApp / Outlook**: Quick communication
- **Trello / Notion**: Task Tracking
- **GitHub**: Code version control

5. Communication Effectiveness:

Conduct structured meetings to enhance focus and participation; share agendas in advance to reduce confusion; encourage open discussion to limit bias; and use both real-time and asynchronous tools to maintain accessibility.

Project Selection Matrix:

Criteria	Weight	ML (Dropout Prediction)	Enterprise Architecture	Process Mining	AI Chatbot
Strategic alignment	25%	5	4	3	4
Feasibility	20%	4	3	3	4
Expected benefit to the organisation	20%	5	4	3	3
Cost within budget	15%	4	2	4	3
Risk level	10%	4	3	4	3
Resource availability	10%	5	3	3	4
Weighted Total Score	100%	**4.65**	**3.30**	**3.20**	**3.60**

Scoring guide: Rate each project option from 1(lowest) to 5(highest) against the criterion.

The highest total score indicates the most suitable project. Risk is scored where 5 = low risk and 1 = high risk. Weighted score = score × weight, summed across all criteria.

Result:

From the concluded matrix, the results indicate that the Machine Learning Dropout Prediction system is selected, because it scored the highest across all the weighted criteria.

Our Teams available skills lie within Python and Data analysis, tools that will be used are free and accessible, and the key importance is that the project addresses a real urgent problem, **student dropout.**

This aligns directly with the institution's strategic goals. The benefit is the model's accuracy and the estimated cost within the group's proposed budget, since most of the tools are open-source and publicly available.

The risk is mostly manageable given our team's technical background and their given skill set.

With the selection process, it follows the OPM401 steps of recommendation:

- The need - student dropout is a rapidly growing problem.
- Potential project listed
- Evaluation criteria defined
- Each option was given a score
- Final decision made by combining the scores and our team's collective and decisive judgment.

Project Proposal:

Project Title: Machine Learning Student Dropout Prediction System
Submitted by: C.A.R.M.B (Creative And Resilient Minds Bonding) – OPM401 Advanced Diploma, NMU
Date: 8 May 2026
Submitted to: Dr. T. Seaba

Objective:
To design, develop, and deploy a machine learning model that predicts the likelihood of student dropout at Nelson Mandela University. The system is set to achieve a target accuracy of at least 85% by analysing historical data and student past/present trends. The project will be completed within a 6-month turnaround and within a maximum budget of R15,000.

Scope:
The project includes data collection and cleaning, data analysis, model development and training, model evaluation, and deployment of a basic user interface for prediction output.
The project excludes full integration with institutional databases, development of a complete student management system, and long-term system maintenance beyond project delivery.

Expected Benefits:
- Early identification of at-risk students
- Improved student retention and success rates
- Enhanced data-driven decision-making for academic interventions
- Cost-effective solution using open-source technologies

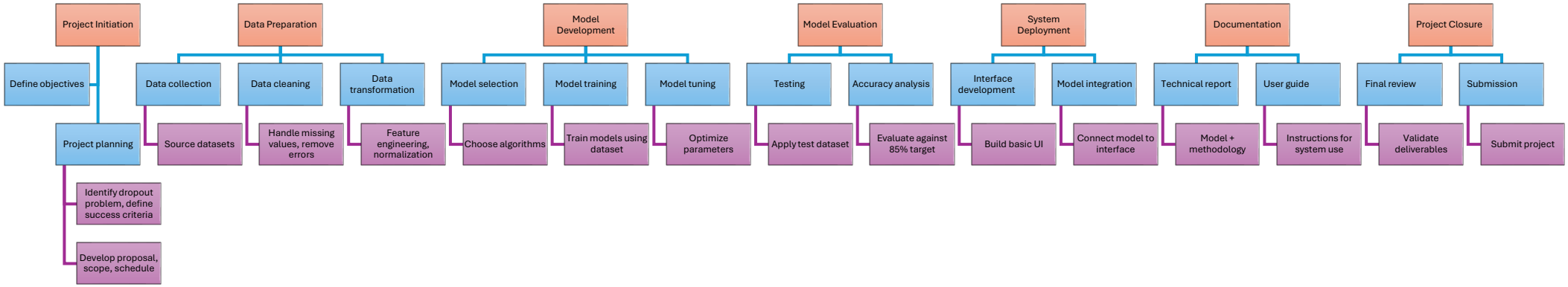
Deliverables	Timeline
Data collection and cleaning report	Week 4
Trained and evaluated a machine learning model	Week 12
Model evaluation and accuracy report	Week 14
Deployed prediction system (basic interface)	Week 20
Final documentation and user guide	Week 24

Budget Estimate:	Estimated Cost
Cloud computing resources	R2,000
Dataset sourcing/licensing	R1,500
Software tools and licenses	R1,000
Documentation and printing	R500
Contingency (approx. 10%)	R500
	Total Estimated Cost: R5,500

The estimated cost is significantly below the allocated budget of R15,000, allowing flexibility for unforeseen expenses or future enhancements.

Feasibility Assessment:
The project is technically feasible as the team possesses the necessary skills in Python programming and data analysis required for machine learning development. It is economically feasible due to the low estimated cost and reliance on open-source tools. Operational feasibility is supported by the 6-month timeline, which is achievable with effective time management.
However, potential risks include data quality limitations, variability in model accuracy, and time constraints due to academic commitments. These risks will be mitigated through thorough data preprocessing, iterative model evaluation, and structured project scheduling.

Scope, Quality, WBS & RAM:



Task	Project Manager	Data Engineer	ML Developer	Software Developer	QA Tester
Project Planning	R	A	A	A	A
Data Collection	A	R	C	C	C
Data Cleaning	A	R	C	C	C
Model Development	C	C	R	A	C
Model Evaluation	C	S	R	C	R
System Development	C	C	A	R	C
Testing	A	C	C	C	R
Documentation	R	A	A	A	A

Legend:

R = Responsible
A = Accountable
C = Consulted

The WBS ensures that the project is broken down into manageable and measurable components, reducing complexity and improving control. By structuring the project into phases such as data preparation, model development, and deployment, the team can track progress effectively and identify delays early.

The RAM enhances accountability by clearly assigning responsibility to each team member based on their expertise. For example, the Data Engineer is responsible for data preparation tasks, while the ML Developer focuses on model development. This prevents role ambiguity, which is a common cause of project failure.

However, some overlap exists - particularly between the ML Developer and QA Tester during model evaluation. This is intentional, as collaborative validation improves model reliability and reduces the risk of bias or inaccurate predictions.

Quality planning ensures that the project not only focuses on completion but also on performance standards. The 85% accuracy requirement aligns with the project objective and provides a measurable benchmark for success.

Project Scheduling:

This involves getting things done on time. It involves putting project activities in order by estimating how long each task will take. The Critical Path Method is important, as this path directly affects how long the whole project will take. This also ensures that the project will finish on time.

No.	Activity	Response.	Est. Dur.	ES	EF	LS	LF	Slack	Predecessor
A	Project Planning	Rohit	2	0	2	0	2	0	-
B	Data Collection	Amity	2	2	4	2	4	0	A
C	Data Cleaning	Amity	3	4	7	4	7	0	B
D	Data Transformation	Bantu	2	7	9	7	9	0	C
E	Model Development	Cameron	4	9	13	9	13	0	D
F	Model Evaluation	Marcellos	2	13	15	13	16	1	E
G	System Development	Rohit	3	13	16	14	16	0	E
H	Testing & QA	Bantu	2	16	18	16	18	0	F, G
I	Documentation	Cameron	2	18	20	18	20	0	H
J	Final Submission	Marcellos	1	20	21	20	21	0	I

Critical Path: A-B-C-D-E-G-H-I-J



Project Duration: 21 Weeks

The schedule above shows an order of activities that follow the machine learning development lifecycle. The data preparation phases are done first because they are very important for how the model works and for reducing mistakes later. The critical path shows that system development and model development are important parts. If there are delays in these areas, it will affect when the project is finished. So, we need to watch these activities and make sure we have enough resources for them. We have some time for model evaluation, which means we can make changes and improvements without slowing down the whole project. This is especially important for machine learning projects because we may need to try a few times to get the model to work correctly and meet our goal of 85%. The schedule also shows that some things can be done at the same time. For example, system development and model evaluation can happen together. This saves time. We need to make sure everyone is communicating well, especially the ML Developer and the Software Developer. Overall, using the Critical Path Method helps us find the important activities to use our resources wisely and make sure the project is finished within 6 months.

Resource Requirements Plan:

Resource planning ensures the right people, tools, and materials are available at the right time to support project activities. According to Gido, Clements, and Baker (Jack Gido, 2017). Resource levelling is a technique used to even out resource usage and prevent overallocation, without unnecessarily extending the project schedule. Resources include people, materials, equipment, facilities, and tools — all of which must be estimated by type and quantity per activity. Effective resource planning controls costs, supports schedule adherence, and reduces the risk of team burnout throughout the project lifecycle. (Schwalbe, 2019)

1. Resource Requirements Table:

Activity	Resources Required	Qty	Duration	Allocation / Resource Levelling Notes
Data Collection & Preparation	Data Engineer, ML Developer	2	3 weeks	Full-time - sourcing NMU student datasets, cleaning, and feature engineering
Exploratory Data Analysis (EDA)	ML Developer, Data Engineer	2	2 weeks	Full-time - statistical analysis and visualisation of dropout patterns
Model Development & Training	ML Developers	2	4 weeks	Full-time - building and training classification models (Python/Scikit-learn)
Model Evaluation & Tuning	ML Developer, QA Tester	2	3 weeks	Full-time - cross-validation, hyperparameter tuning; target 85% accuracy
Backend API Development	ML Developer	1	2 weeks	Full-time - Flask/FastAPI endpoint; levelled to start after model sign-off
Testing & Quality Assurance	QA Tester	1	2 weeks	Full-time - unit, integration, and UAT; runs Weeks 17–18 post-development
Deployment	ML Developer, Data Engineer	2	1 week	Full-time - Google Colab Pro/cloud deployment; Week 19 only
Documentation & Reporting	All team members	5	2 weeks	Part-time (50%) - final report, user guide, lessons learned; Weeks 20–21
Project Management	Project Manager (Rohit Aroonslam)	1	24 weeks	50% throughout - planning, coordination, risk monitoring, stakeholder reporting

2. Resource Levelling Discussion:

Resource levelling was applied to balance the team workload across the 24-week project schedule. The most significant peak occurs during Weeks 11–16 (Model Development and Training), where two ML Developers work concurrently at full capacity. To reduce pressure during this phase, the Backend API Development task was deliberately delayed, beginning only after model sign-off, preventing a concurrent overload of the same developers. Similarly, the QA Tester is introduced only in Weeks 17–18, maintaining a sequential dependency that prevents idle time earlier in the schedule. The Project Manager operates at 50% allocation throughout the entire project to provide continuous oversight without becoming a resource bottleneck. Any scope additions must be processed through a formal change control process to prevent unplanned resource demand. (Jack Gido, 2017) (Schwalbe, 2019).

Costs, Budgets, and Earned Value:

Cost management is the process of estimating, budgeting, and controlling costs to ensure the project is within plan. A time-phased budget allocates the costs throughout the project schedule. The performance measures of Earned Value Management (EVM) are Planned Value (PV, planned cost), Earned Value (EV, completed work value), and Actual Cost (AC, actual spending). Additional evaluation of project performance is done through Cost Performance Index (CPI) and Schedule Performance Index (SPI). At Week 14, the project is on budget (CPI 1.07) but slightly behind schedule (SPI 0.94), and a few schedule adjustments are necessary.

Total Project Budget

Budget Item	Total Cost
Human resources (team time value)	R8,000
Google Colab Pro (cloud computing)	R2,000
Dataset sourcing	R1,500
Software and tools	R500
Documentation and printing	R500
Contingency (10%)	R1,250
Total Project Budget	**R13,750**

Time-Phased Budget (Planned Value by Phase):

Phase	Weeks	Budget Allocated	Cumulative PV
Planning	1–1	R1,000	R1,000
Data Management	2–10	R3,500	R4,500
Model Development	11–16	R4,000	R8,500
Testing & QA	17–19	R2,000	R10,500
Deployment	20–21	R1,500	R12,000
Documentation & Closure	22–24	R1,750	R13,750
Phase	**Weeks**	**Budget Allocated**	**Cumulative PV**

Time-Phased Budget (Planned Value by Phase):

Assume the project is currently at **Week 14** (end of the Model Training phase)

EVM Metric	Formula	Value	Interpretation
Planned Value (PV)	Budget planned for Week 14	R8,500	What we planned to spend
Earned Value (EV)	Value of work completed	R8,000	What we achieved
Actual Cost (AC)	Real money spent so far	R7,500	What we have spent
Cost Variance (CV)	EV – AC	+R500	Positive = under budget
Schedule Variance (SV)	EV – PV	-R500	Negative = slightly behind schedule
Cost Performance Index	EV ÷ AC	1.07	Above 1.0 = cost efficient
Schedule Performance Index	EV ÷ PV	0.94	Below 1.0 = slightly behind

Managing Risk:

Risk management is the systematic process of identifying, assessing, and responding to uncertainties that may negatively affect project objectives. (Jack Gido, 2017) Outline five steps: identify risks, analyse likelihood and impact, prioritise, treat or respond, and monitor results. Risk responses include avoidance, mitigation, transfer, and acceptance — with the highest priority given to risks rated both high likelihood and high impact. (PMI, 2025)

1. Risk Assessment Matrix (Figure 8.1):

Risk	Impact	Likelihood	Impact	Action on Trigger	Responsibility	Response Plan
Scope creep	Increased cost, delayed delivery	H	H	Scope change submitted	Project Manager	Apply change control; re-baseline scope, schedule & budget; notify supervisor
Data quality issues	Model accuracy below 85% target	H	H	Model performance drops below 80%	ML Developer	Validate dataset early; apply data cleaning pipeline; source additional data.
Developer unavailability	Milestone delays, knowledge loss	M	H	Team member absent >3 days	Project Manager	Cross-train team; always maintain shared repo and documentation
Budget overrun	Reduced tools, sponsor dissatisfaction	M	M	CPI drops below 0.9	Project Manager	Track CPI weekly; apply 10% contingency reserve; reduce non-critical spend
Model bias/overfitting	Inaccurate dropout predictions	M	H	Validation accuracy < training acc.	ML Developer	Use cross-validation; apply regularisation; test on holdout dataset
Poor stakeholders buy-in	Low adoption by university admin	M	M	The stakeholder missed two reviews	Project Manager	Schedule early demos; provide bi-weekly progress reports to the supervisor.

2. Risk Register:

ID	Date Raised	Risk Description	Likelihood	Impact	Severity Score	Owner	Mitigating Action
R1	05/05/26	Scope not well-defined; requirements may expand mid-project	Likely	High	16	Project Sponsor	Complete project charter; lock scope after sign-off; apply change control
R2	05/05/26	Training data insufficient or too noisy to reach 85% accuracy target	Likely	High	16	ML Developer	Validate data quality in Week 1; source additional datasets if needed
R3	05/05/26	Key ML developer becomes unavailable due to illness or withdrawal	Possible	High	12	Project Manager	Cross-train team; maintain shared GitHub repo and documentation
R5	05/05/26	Budget overrun due to cloud computing costs exceeding Colab Pro	Possible	Medium	9	Project Manager	Track spend weekly; apply 10% contingency reserve

Severity Rating = Likelihood × Impact. Green ≤4 (Low) | Yellow 5–8 (Medium) | Orange 9–15 (High) | Red >15 (Critical). Critical/High risks escalated immediately to the project supervisor. (PMI, 2025) (Jack Gido, 2017)

Closing the Project:

Project closure is the final phase of the project management lifecycle, ensuring all activities are formally completed, deliverables accepted by the client, and resources released.

Category	Criteria	Achieved (Y/N)
Objectives	To design, develop, and deploy a machine learning model that predicts the likelihood of student dropout at Nelson Mandela University.	Y
Benefits	• Early identification of at-risk students • Improved student retention and success rates • Enhanced data-driven decision-making for academic interventions • Cost-effective solution using open-source technologies	Y
Deliverables	• Data collection and cleaning report • Trained and evaluated the machine learning model • Model evaluation and accuracy report • Deployed prediction system (basic interface) • Final documentation and user guide	Y

Lessons Learned:

The team learned the value of effective collaboration between technical team members, academic staff, and project stakeholders. Continuous testing played a crucial role in the testing phase before launching or moving to the next phase, as it helped pinpoint problems, minor inaccuracies, and how accurate the probability data was, given how much or little information/data was provided for the system to give its probability prediction. The data used was on past students without knowing the outcome.

Stakeholder feedback:

Academic staff were mighty impressed with the results and accurate predictions based on the information/data provided to predict future test results as well as dropout/failure probability. The staff plans to use this model in the near future to predict who may fail, which may help the staff to give extra support and lessons for those currently struggling.

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